

D801

Machine Learning for Computer Scientists

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A retailer has purchase histories for 2 million customers but no segment labels. The marketing team wants to discover natural customer groups it has not defined yet. The data sits in a cloud warehouse, and the marketing director has a \$300,000 budget. Which approach fits?

- A. Machine learning always needs labels, so hand-label all 2 million customers first.
- B. A clustering method such as k-means on scaled purchase features.
- C. Linear regression on total spend.
- D. Logistic regression to predict each customer's segment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A streaming service wants to predict whether each subscriber will cancel next month (yes or no). It has 14 months of usage data per subscriber, and the team works in scikit-learn. Which algorithm fits the target?

- A. Linear regression on the 14 monthly usage totals
- B. k-means clustering
- C. Logistic regression
- D. Linear regression

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A decision tree with no depth limit scores 99% accuracy on training data and 71% on validation data. It was trained on a GPU instance using 18 features. What is the best next step?

- A. Allow even more depth so it learns the data better.
- B. Move to a larger GPU instance.
- C. Report 99% as the expected accuracy in production.
- D. Limit tree depth or prune it, since the model is overfitting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A linear model scores 62% on training data and 60% on validation data; the target is 85%. What is the likely problem and fix?

- A.** Overfitting; add stronger regularization.
- B.** Underfitting; add informative features or use a more flexible model.
- C.** Collect more rows of the same data.
- D.** Nothing, because matching scores always mean a good model.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A team standardizes all 50,000 rows, fitting the scaler on the full dataset, and then splits the data 80/20 into training and test sets. The data came from a Postgres warehouse and has 30 features. Test accuracy looks surprisingly high. What is wrong?

- A.** The Postgres export probably corrupted the data.
- B.** The split should be 50/50.
- C.** Nothing; scaling does not change labels, so it cannot leak.
- D.** Test-set statistics leaked into training; fit the scaler on the training set only and apply it to the test set.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A lender has 1,200 labeled loan records collected from 3 branches and stored as Parquet files. It runs 5-fold cross-validation. In each fold, how many records are used to train the model?

- A.** 960
- B.** 1,200
- C.** 400
- D.** 240

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In a k-fold cross-validation, each fold validates on 150 records and trains on 1,050. What is k?

- A.** 150
- B.** 10, because k is always 10.
- C.** 7
- D.** 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A fraud model scores 10,000 transactions, of which 100 are actually fraud. It flags 80 transactions, and 60 of those are fraud. The model runs on a managed cloud endpoint and was retrained over a weekend. What share of actual fraud did it catch?

- A. 60%
- B. 75%
- C. 99.4%
- D. 0.8%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A fraud team wants fewer honest customers wrongly blocked, even if a few more fraud cases slip through. Which metric should it raise?

- A. Accuracy
- B. Recall, since raising recall always raises precision too.
- C. Precision
- D. Recall

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A team is tuning the regularization setting C for a support vector machine and has 10 candidate values. Training runs on 4 vCPUs, and the data has 25 features plus a held-out test set. How should it choose C?

- A. Let the SVM learn C during fitting, like its weights.
- B. Pick the C with the best test-set score.
- C. Pick the C with the best cross-validated score on the training data, then check the test set once.
- D. Pick the C with the best training accuracy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A single deep decision tree is unstable: small changes in the training data produce very different trees and predictions. The team has 16 CPU cores and a three-day deadline. Which ensemble most directly addresses the problem?

- A. Stacking, because it is always the most accurate ensemble.
- B. Bagging, such as a random forest that averages many trees built on bootstrap samples.
- C. Boosting shallow trees in sequence.
- D. A single even deeper tree.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

A shallow model underfits. The team wants models trained one after another, each focusing on the previous model's errors. Which technique is this?

- A. Boosting
- B. Cross-validation
- C. Bagging
- D. Pruning each tree harder

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A retailer has purchase histories for 2 million customers but no segment labels. The marketing team wants to discover natural customer groups it has not defined yet. ~~The data sits in a cloud warehouse, and the marketing director has a \$300,000 budget.~~ Which approach fits?

Target word: method family for unlabeled grouping **Decides:** No labels exist; the goal is to discover groups.

A	Absolute language	Unsupervised learning works without labels.
B	Correct	Unsupervised clustering finds groups without labels.
C	Confusion set	Predicts a number from labeled examples; it does not form groups.
D	Adjacent category	A supervised classifier needs segment labels that do not exist.

2. Answer C Concept 2

A streaming service wants to predict whether each subscriber will cancel next month (yes or no). ~~It has 14 months of usage data per subscriber, and the team works in scikit learn.~~ Which algorithm fits the target?

Target word: model type for a yes/no target **Decides:** The outcome is a category (cancel or not).

A	Camouflage	The amount of history does not change the target type.
B	Adjacent category	Unsupervised; it groups subscribers but does not predict the labeled outcome.
C	Correct	A classifier that models the probability of a binary outcome.
D	Confusion set	Predicts an unbounded number; the shared word “regression” is the trap.

3. Answer D Concept 3

A decision tree with no depth limit scores 99% accuracy on training data and 71% on validation data. ~~It was trained on a GPU instance using 18 features.~~ What is the best next step?

Target word: diagnosis and fix **Decides:** Large gap between training and validation scores.

A	Inverted operation	More complexity widens the overfit gap.
B	Camouflage	Hardware does not change how well the model generalizes.
C	Confusion set	Training accuracy is not an estimate of performance on new data.
D	Correct	Constraining complexity reduces variance and narrows the gap.

4. Answer B Concept 3 · transfer

A linear model scores 62% on training data and 60% on validation data; the target is 85%. What is the likely problem and fix?

Target word: reverse: diagnose underfitting **Decides:** Both scores are low and close together.

A	Inverted operation	Regularization adds bias, making an underfit model worse.
B	Correct	Low scores on both sets signal high bias.
C	Adjacent category	More data mainly helps variance, not a model too simple for the pattern.
D	Absolute language	Matching scores rule out overfitting, not underfitting.

5. Answer D Concept 4

A team standardizes all 50,000 rows, fitting the scaler on the full dataset, and then splits the data 80/20 into training and test sets. ~~The data came from a Postgres warehouse and has 30 features.~~ Test accuracy looks surprisingly high. What is wrong?

Target word: what flaw inflates the test score **Decides:** Scaler was fit before the split.

A	Camouflage	The data source does not decide the leakage.
B	Adjacent category	Split ratio is not the flaw; the order of steps is.
C	Dropped element	Leakage comes from the test set's means and spreads, not labels.
D	Correct	Preprocessing fit on all rows lets test data shape the model.

6. Answer A Concept 5

A lender has 1,200 labeled loan records ~~collected from 3 branches and stored as Parquet files.~~ It runs 5-fold cross-validation. In each fold, how many records are used to train the model?

Target word: training rows per fold **Decides:** Each fold trains on 4/5 of the data.

A	Correct	$1,200 \times 4/5 = 960$; the other 240 validate.
B	Dropped element	Ignores the held-out fold.
C	Camouflage	$1,200 \div 3$ uses the branch count, which does not decide folds.
D	Inverted operation	That is the validation fold, not the training data.

7. Answer D Concept 5 · transfer

In a k-fold cross-validation, each fold validates on 150 records and trains on 1,050. What is k?

Target word: solve for the number of folds **Decides:** Total = 1,200; $k = 1,200 \div 150$.

A	Wrong stem number	The fold size, not the number of folds.
B	Absolute language	10 is common, but k follows from the data here.
C	Dropped element	$1,050 \div 150$ leaves out the validation fold.
D	Correct	$1,200 \div 150 = 8$ folds.

8. Answer A Concept 6

A fraud model scores 10,000 transactions, of which 100 are actually fraud. It flags 80 transactions, and 60 of those are fraud. ~~The model runs on a managed cloud endpoint and was retrained over a weekend.~~ What share of actual fraud did it catch?

Target word: recall **Decides:** $\text{Caught} \div \text{actual fraud} = 60 \div 100$.

A	Correct	Recall = $60 \div 100$.
B	Confusion set	$60 \div 80$ is precision — the share of flags that were right.
C	Adjacent category	Accuracy, $(60 + 9,880) \div 10,000$, looks high because fraud is rare.
D	Wrong stem number	$80 \div 10,000$ is the flag rate.

9. Answer C Concept 6 · transfer

A fraud team wants fewer honest customers wrongly blocked, even if a few more fraud cases slip through. Which metric should it raise?

Target word: reverse: from the business goal to the metric **Decides:** Wrong blocks are false positives.

A	Adjacent category	Dominated by the huge honest class; hides the trade-off.
B	Absolute language	They usually trade off against each other.
C	Correct	Precision rises as false positives fall.
D	Inverted operation	Recall targets missed fraud (false negatives), the opposite trade-off.

10. Answer C Concept 7

A team is tuning the regularization setting C for a support vector machine and has 10 candidate values. ~~Training runs on 4 vCPUs, and the data has 25 features~~ plus a held-out test set. How should it choose C?

Target word: correct tuning procedure **Decides:** C is a hyperparameter; the test set must stay untouched.

A	Confusion set	C is set before training; weights are the learned parameters.
B	Inverted operation	Tuning on the test set makes the final estimate too optimistic.
C	Correct	Validation picks hyperparameters; the test set gives an unbiased final estimate.
D	Dropped element	Leaves out validation, so it favors the most overfit setting.

11. Answer B Concept 8

A single deep decision tree is unstable: small changes in the training data produce very different trees and predictions. ~~The team has 16 CPU cores and a three-day deadline.~~ Which ensemble most directly addresses the problem?

Target word: ensemble for high variance **Decides:** The problem is instability (variance).

A	Absolute language	No ensemble is always best; the variance problem points to bagging.
B	Correct	Averaging independent models cuts variance.
C	Confusion set	Boosting mainly reduces bias; it can add variance if overdone.
D	Inverted operation	More depth increases variance.

12. Answer A Concept 8 · transfer

A shallow model underfits. The team wants models trained one after another, each focusing on the previous model's errors. Which technique is this?

Target word: reverse: from the description to the ensemble **Decides:** Sequential training on prior errors.

A	Correct	Boosting builds models in sequence to reduce bias.
B	Adjacent category	An evaluation method, not an ensemble.
C	Confusion set	Trains models independently in parallel.
D	Inverted operation	Makes each model simpler, worsening underfitting.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Supervised vs. unsupervised learning	Supervised learning needs labeled targets; unsupervised methods such as clustering and dimensionality reduction find structure in unlabeled data.	Reaching for a classifier when no labels exist.	“No labels,” “discover groups,” or “segments we don’t know yet.”
2	Regression vs. classification	Predicting a number is regression; predicting a category is classification — and logistic regression is a classifier.	Choosing linear regression because the word “regression” appears.	Yes/no or category outcome vs. a continuous amount.
3	Overfitting and underfitting	High training score with a much lower validation score means overfitting (high variance); low scores on both mean underfitting (high bias).	Adding complexity to an overfit model or regularizing an underfit one.	A gap between training and validation scores, or two low scores.
4	Data leakage	Fit every preprocessing step (scaling, imputation, encoding) on training data only, then apply it to validation and test data.	Transforming the full dataset before splitting.	A scaler or imputer fit before the train/test split.
5	k-fold cross-validation	With k folds, each fold is the validation set once; each model trains on $(k - 1)/k$ of the data.	Mixing up the training share with the validation share.	A row count and a number of folds.
6	Precision, recall, and accuracy	Precision = $TP \div \text{predicted positives}$; recall = $TP \div \text{actual positives}$; on imbalanced data accuracy can look high while the model misses positives.	Reporting accuracy or swapping precision and recall.	A rare positive class and a question about how many were caught or how many flags were right.
7	Hyperparameter tuning	Hyperparameters are set before training and chosen with validation or cross-validation; the test set is used once at the end.	Tuning on the test set or picking the setting with the best training score.	Several candidate values for a setting like C, depth, or learning rate.
8	Bagging vs. boosting	Bagging (e.g., random forest) averages independently trained models to cut variance; boosting trains models in sequence on prior errors to cut bias.	Using boosting to fix an unstable model or bagging to fix an underfit one.	“Unstable” or “high variance” vs. “too simple” or “sequential.”

Course scope (public description): Machine Learning for Computer Scientists provides a comprehensive introduction to foundational machine learning algorithms and their applications, aligned with the AWS Certified Machine Learning Engineer — Associate (MLA-C01). The course covers supervised learning techniques such as linear regression, logistic regression, decision trees, and support vector machines, as well as unsupervised learning methods like clustering and dimensionality reduction. Essential concepts in model evaluation, selection, and optimization are also covered, including cross validation, hyperparameter tuning, and ensemble methods. Designed specifically for computer scientists, this course sets the groundwork for advanced studies in deep learning and natural language processing. This course teaches students machine learning (ML) techniques of

supervised and unsupervised ML methods using Python programming language.